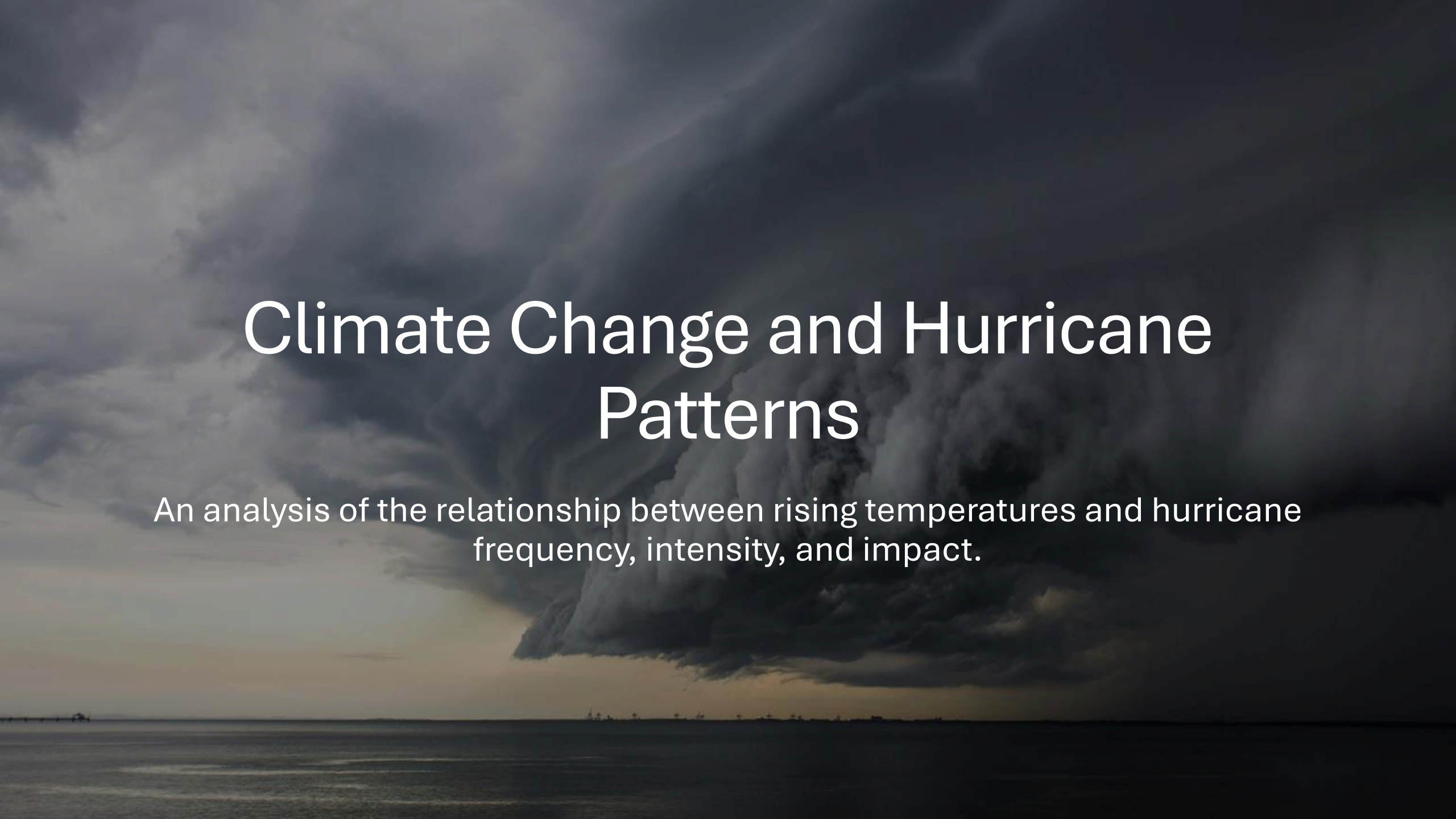




Climate Change and Hurricane Patterns

An analysis of the relationship between rising temperatures and hurricane frequency, intensity, and impact.

Objective

- This presentation examines how climate change, specifically rising temperatures, correlates with the frequency, intensity, and impact of hurricanes over time.

- Shaya Engelman
- Data 608 Story 5



Background

- Hurricanes, and other tropical storms, cause tremendous damage. They cause millions of dollars in damage, take people's lives, make people lose their homes and possessions, and leave the survivors to deal with the damage.
- Scientists have observed an increase in these storms as temperatures increase.
- Global temperatures have been steadily increasing with time.
- The results of these two effects are dire.

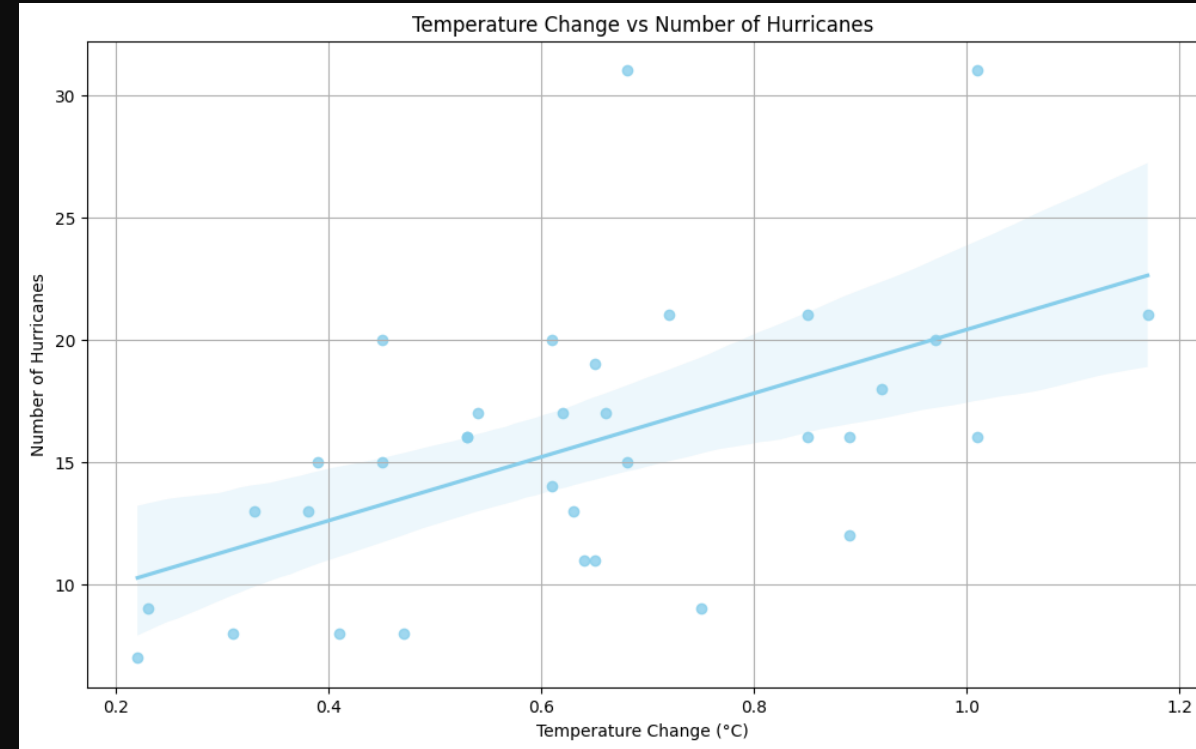


Hurricane Ian (2022) Devastation

- In 2022, Florida was devastated by hurricane Ian. The damage it caused destroyed people's lives. It caused 161 fatalities and an estimated \$113 billion in damages.

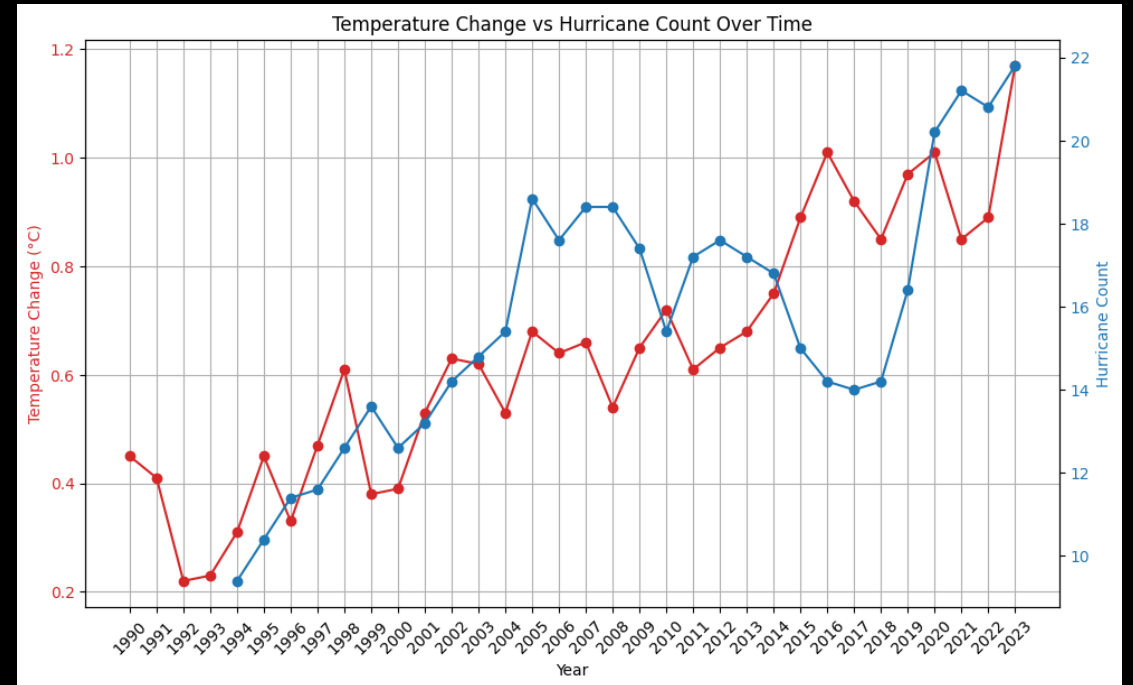
Temperature Change vs. Number of Hurricanes

- Key Insight:
 - As temperature increases, the number of hurricanes tends to rise, suggesting a potential climate-driven increase in hurricane activity.
 - This suggests making efforts to keep temperatures from rising is very prudent



Temperature Change and Hurricane Count Over Time

- Over the last 30+ years, both temperatures and total hurricane counts have steadily increased
- A rolling average of five years was used for the hurricane count to smooth it out a bit and help minimize the effects of random fluctuations
- The positive linear relationship is clear in both lines but due to the low overall count of hurricanes, it isn't reflected in every given year even with the smoothing applied





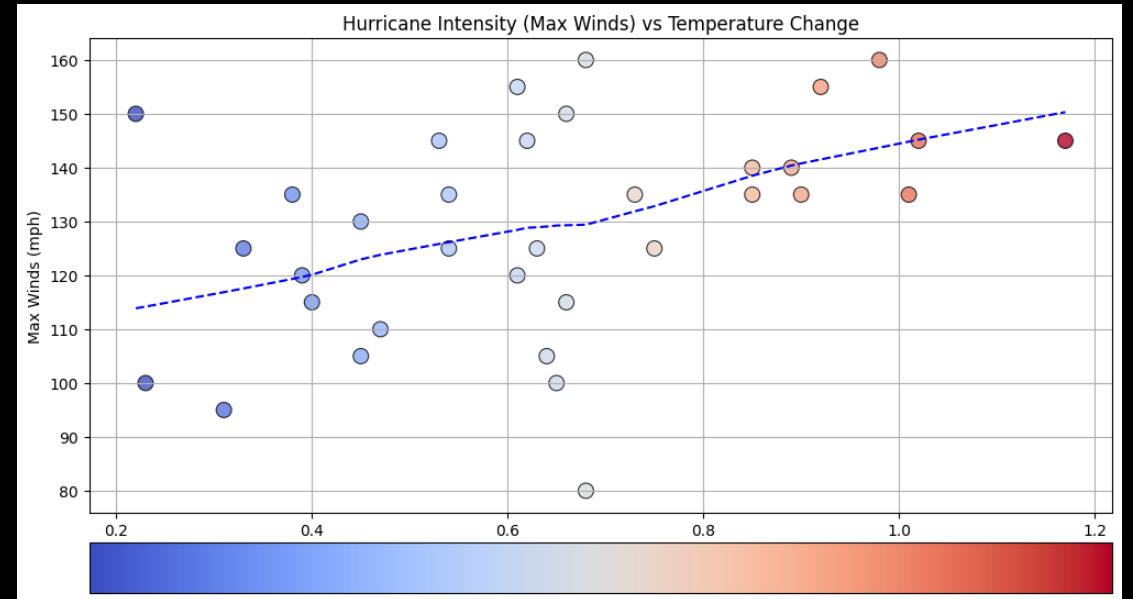
Hurricane Intensity

Besides for greater temperatures causing an increase in the total amount of hurricanes, there also seems to be stronger hurricanes than before.



Hurricane Intensity (Max Winds) vs Temperature Change

- In addition to total hurricane counts rising with temperature, the intensity of hurricanes do as well.
- This plot illustrates the rising maximum intensity of hurricanes with rising temperatures
- Redder dots represent warmer temperatures and bluer dots warmer ones.



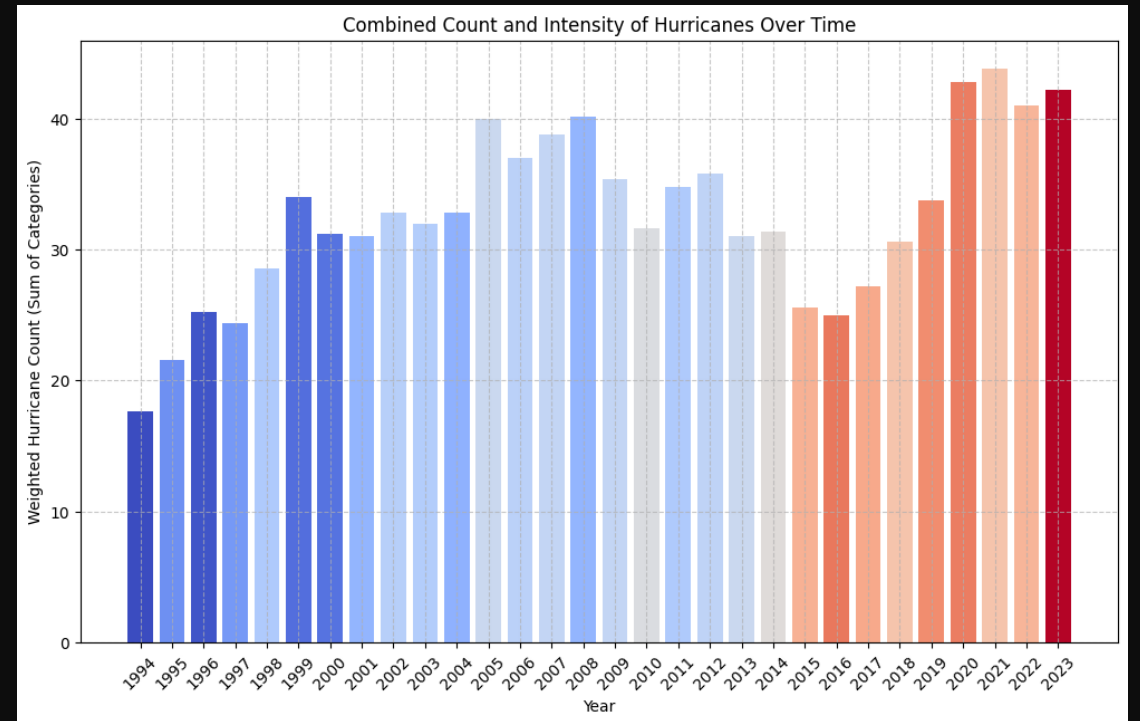
Putting It All Together

- Increases temperature is clearly correlated with increased frequency of hurricanes
 - Similarly, increased temperature is correlated with more intense hurricanes
 - Global temperatures have been steadily increasing in recent memory
 - This results in an increase in both frequency and intensity of storms over time
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As Global Temperatures Increase, Hurricane Frequency and Intensity Increase

- Ranking Hurricanes:
 - We ranked hurricanes from 1 (weakest) to 5 (strongest) on a scale called the Saffir-Simpson Hurricane Wind Scale.
- Yearly Scores:
 - To see how hurricane strength and numbers change over the years, we add up scores each year. Stronger hurricanes contribute more points.
- Bar Graph:
 - The bars show these yearly scores, averaged over five years to smooth out any odd spikes and make the trend clearer.
- Temperature Trend (Red Line):
 - The red line represents how Earth's temperature has changed over the same years.
- Key Insight:
 - By looking at both the bars and the red line, we can explore if rising temperatures lead to stronger hurricanes.



Takeaway

- We are experiencing a dramatic rise in both the frequency and intensity of hurricanes, and these powerful storms are wreaking havoc on our communities. In these challenging times, it's heartening to see people coming together to help each other recover from the aftermath of such disasters. However, the true challenge lies in motivating individuals to change their habits to prevent these events in the first place.
 - Often, the effort required to reduce or prevent these disasters is significantly less than the effort needed to rebuild after they strike. Imagine the impact we could make if everyone took small steps today to protect our planet. By understanding these findings, we can inspire each other to take action and make a difference, ensuring a safer and more sustainable future for all. Together, we have the power to prevent these catastrophes from happening, rather than just reacting to them. Let's commit to creating a better world, starting now.
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Data Sources

- Global Temperature Data Sourced From Here: [National Aeronautics and Space Administration](#)
- Hurricane Data Sourced From Here: [National Hurricane Center and Central Pacific Hurricane Center](#)

